

Q1. Data Mining -

Data Mining is the process of discovering meaningful patterns, correlations and knowledge from large datasets using statistical, machine learning and database management techniques.

STEPS in KDD Process:-

1. Data Cleaning - Remove noise, inconsistencies and missing values
2. Data Integration - Combines data from multiple sources into a coherent store.
3. Data Selection - Extracts relevant data for analysis.
4. Data Transformation - Converts data into suitable formats (eg-normalization, aggregation)
5. Data Mining - Applies algorithms to extract useful patterns and relationships.
6. Pattern Evaluation - Identifies interesting, valid, novel and useful patterns.
7. Knowledge Presentation - Represents mined knowledge using visualization and reports.

Diag

Data Cleaning



Data Integration



Data Selection

→ Data Transformation →

Data Mining

Knowledge Presentation

↑
Pattern Evaluation
↑

Q2. Applications -

- Business - Customer Segmentation, sales forecasting, fraud detection and credit risk analysis.
- Marketing - Targeted advertising, market basis analysis, customer churn prediction, recommendation systems.
- Research - Identifying trends in scientific data, analyzing experimental results and optimizing research outcomes.

Contribution to Process Improvement -

- Enhances decision-making through data-driven insights
- Reduces operational costs by identifying inefficiencies
- Improves customer satisfaction via personalization
- Detects ~~errors~~ anomalies early to prevent losses
- Optimizes resource allocation and workflow automation.

Q3. Classification

A supervised learning technique that categorizes data into predefined labels or classes.

Example - Classifying emails as spam/ham

Prediction

Estimates future or unknown values based on current patterns and historical data.

Example - Predicting next month's sales revenue.

<u>Basis</u>	<u>CLASSIFICATION</u>	<u>PREDICTION</u>
Output Type	Categorical	Continuous
Goal	Identifying class label	Forecast numeric value
Example	Loan Approval (Yes/No)	Predicting credit score
Algorithms	Decision Tree, Naive Bayes	Regression, Neural Networks

Q4. Working of Decision Tree Induction:

1. Select the best attribute using metrics like Information Gain or Gini Index.

2. Split the dataset into subsets based on the chosen attribute's values.
3. Repeat recursively for each subset until nodes become homogeneous or meet a stopping criterion
4. The resulting structure is a tree with decision ~~tree~~ nodes and leaf nodes representing outcomes.

Example -

To decide whether to play tennis:

- Attribute: weather
- If sunny → check Humidity
- If rainy → check Wind
- If overcast → Play → yes

Key Issues -

- ~~Overfitting or underfitting nodes~~
- Handling missing or noisy data
- Choosing the right attribute selection measure.
- Scalability with large datasets.
- Interpretability and accuracy trade-offs.

Q5.

KNN algorithm

- A lazy learning method that classifies a new instance based on the majority vote of its k nearest neighbours.
- Distances metrics like Euclidean or Manhattan are used to find similarity.

Steps :-

1. Choose the number of neighbours k
2. Computes distances between the new sample and existing samples.
3. Select k nearest points
4. Assign the class most common among those neighbours.

Advantages -

- Simple and Effective
- No training phase required
- Works well for small datasets

Limitation -

- Computationally expensive for large datasets
- Sensitive to noise and irrelevant features
- Choice of k greatly affects performance.

Applications

- Pattern and image recognition
- Recommendation systems
- Credit scoring and fraud detection

Q6.Benefits

- Supports strategic decision-making
- Enhances customer retention and satisfaction
- Detects fraud and anomalies early
- Increases operational efficiency
- Helps in predictive analysis for future planning.

Challenges -

- Data privacy and security concerns
- Integration with existing systems
- Need for skilled personnel
- High computational and storage requirements
- Difficulty in interpreting and complex models.